Apoorv **Kothari**

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Summary

Experienced Rust developer with 15+ years of software development expertise, five of them focused on transport libraries used to serve billions of requests per day. Past experiences include releasing a production-ready cryptographic transport library, deploying services serving millions of requests/sec, migrating entire microservice backend to Kubernetes, and delivering complex Android solutions. Expert in Rust, QUIC, TLS, backend development and scalable software solutions. I am passionate about privacy and security.

Skills and Contributions

Skills: QUIC, TLS, Protocols, Cryptography, Backend Engineering, Android, and distributed systems. Rust, C, Java, Go.
Open source contributions: s2n-quic, s2n-tls, quiche, boring

Work Experience

Cloudflare — *Remote*

Tech Lead / Systems Engineer - Protocols — *Feb. 2025 - May 2026*

- Led an org-critical migration of 5M customers (25M req/sec) from the legacy to the new proxy.
- Deployed Session Resumption and 0-RTT. Used metrics to identify gaps impacting ~13% of resumption attempts.
- Investigated and fixed a bug in our QUIC stack which was incorrectly closing ~2% of connections.
- Fixed security vulnerabilities in our QUIC stack. Coordinated the effort to patch all internal services before public disclosure.
- Root caused and fixed a complex deadlock issue in our Go proxy which was impacting up to 1K+ instances in the fleet.
- Mentored engineers during a risky CDN migration over 1.5 years, enabling the team to move fast while avoiding tech debt.

AWS — *Seattle, WA*

Software Engineer – Cryptography - Transport libraries — *Feb. 2021 - Nov. 2024*

- Core maintainer of s2n-quic and the s2n-tls networking protocol libraries which power the majority of AWS traffic.
- Enabled HTTP/3 support for CloudFront by integrating s2n-quic, resulting in 20% CPU reduction for the networking stack. Collaborated across teams to enable multi-certificate support by adding support for asynchronous certificate loading.
- Built team consensus and implemented core library features by authoring technical documents. Notable accomplishments include ACK Frequency analysis, a zero-cost event framework, and QUIC Connection Migration support.

Software Engineer – EC2 — *Nov. 2019 - Feb. 2021*

- Deployed a critical Fingerprint service to replace 1K+ hardcoded EC2-compatibility rules across 3 services. Developed A/B testing strategy, unit testing, and detailed metrics to achieve migration with zero impact.
- Led the Fingerprint project from RFC to delivery and mentored a junior engineer to take over long-term ownership.
- Owned and operated a core service, repairing 100K+ live customer EC2 hosts/month and prevented production impact.
- Implemented automatic fail-safe switch to reduce blast radius from AWS Regions to AZ, enabling the team to move faster.

iHeartRadio — *New York, NY*

Infrastructure/Backend Software Engineer — *Feb. 2017 - Mar. 2019*

- Led and aligned owners across teams (Data/iOS/Design) and delivered a new music recommendation service, which featured versioned storage of the recommendation data and < 5min rollback capabilities in the case of bad data.
- Migrated production Scala microservice backend from Akka Cluster to Kubernetes over 6 months. The deployment was provisioned by Terraform and paved the path for onboarding other teams (Frontend, Data) in the company.

Consulting — *Remote*

Independent Consultant — *Mar. 2016 - Feb. 2017*

- Built an Android-based, project management solution to simultaneously support office and field personnel at a construction firm. The app featured cloud storage (Google Drive), synchronization, text/image editing.

TouchLab — *New York, NY*

Android Software Engineer — *Sept. 2014 - Jan. 2016*

- Delivered a solution to assist forest rangers with deforestation reporting; featured offline support and custom map rendering.

Education

The Cooper Union for the Advancement of Science and Art *New York, NY*

Bachelor of Engineering | Full-tuition Academic Scholarship Graduation 2012